

US_ID	Feature	User Story	Acceptance Criteria	Story Points
US_01	Image Upload	As an individual user, I want to upload knee X-ray images so that I can receive an AI-based assessment of osteoarthritis severity. +2	The patient can access an interface to upload an image. The patient can select and submit an image from device. The system accepts image files in specified formats: JPEG, JPG, PNG, DICOM. The uploaded image is stored and associated with the patient's case. A confirmation message is displayed upon successful upload.	13
US_02	Image Upload	As a research coordinator, I want to upload multiple X-ray images at once so that I can analyze datasets efficiently. +1	The patient can upload multiple images (up to 3) to one case. The system accepts multiple images in specified formats (JPEG, JPG, PNG). All uploaded images are associated with the same patient case. The patient receives a confirmation message after uploading each image. The system displays a preview of all uploaded images for review before analysis.	8
US_03	User Management	As an individual user, I want to register an account so that I can store my basic information and access personalised services. +2	The user can access a registration page requiring: Full name, Email address, Password creation. Post-registration, the user can log in and update their profile information.	2
US_04	User Management	As an individual user, I want to log in to my account so that I can manage my cases and view diagnoses. +2	The user accesses a login page requiring: Email address, Password. Upon successful login, the user is directed to their dashboard.	1
US_05	User Management	As a user, I want to view my past cases and track their statuses so that I can monitor my health progress. +2	The user can access a 'Case History' section after logging in. All past cases are listed in chronological order (most recent first). Each case displays: case ID, case open date, status. The system updates case statuses in real-time. Clicking on a case entry navigates the user to that case's report.	2
US_06	User Management	As an individual user, I want to implement JWT authentication so that my session is secure. +1	JWT tokens are generated on login and validated on every protected route. Tokens expire after a defined period and require re-authentication. Invalid or expired tokens return a 401 Unauthorized response.	3
US_07	AI Model Integration	As an individual user, I want to receive an OA severity diagnosis so that I know the knee condition I am facing. +2	After submitting the image, the user receives a diagnosis. The diagnosis is presented as the KL grade (0–4) and severity label. The system provides the diagnosis within a reasonable time (< 1 min). The diagnosis is stored and associated with the user's case.	20
US_08	AI Model Integration	As an individual user, I want to receive the most accurate possible diagnosis so that I can trust the assessment provided. +2	The system continuously improves the model and algorithms for better diagnostic results. The diagnosis accuracy is optimised based on model refinements and external validation. The confidence level of the diagnosis should be high (e.g., 95%).	11
US_09	AI Model Integration	As an individual user, I want to receive a brief description of the diagnosed condition so that I can understand the issue I am facing. +2	After receiving the diagnosis, the user is provided with a brief description of the diagnosed condition. The description is written in clear, understandable language. The description includes common symptoms and causes.	2
US_10	AI Model Integration	As an individual user, I want to receive treatment recommendations so that I can take action towards treating my knee condition. +1	After receiving the diagnosis, the user is provided with treatment recommendations. The description is written in clear, understandable language. The description includes common symptoms and causes of the condition.	3